

Data Sharing Practices in Foundational Learning Systems: A Scoping Review of Barriers, Facilitators, Interventions, and Governance Frameworks in Sub-Saharan Africa

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Rigobert Pambe: methodological guidance (JBI methodology and PRISMA-ScR); protocol drafting and review. Azibo Kester Ngwa, Ngom Maceline Anwi, Karkheh Seraphine Ngassah: information retrieval, eligibility screening, and data extraction. All authors reviewed and approved the final protocol. Patrick Okwen (eBASE Africa) will serve as independent adjudicator for unresolved disagreements during screening and extraction.

Abstract

Keywords: foundational learning; data sharing; education management information systems; information governance; institutional trust; Sub-Saharan Africa

Background

Access to reliable and timely educational data is vital for strengthening foundational literacy and numeracy outcomes across Sub-Saharan Africa (SSA). Despite growing investment in Education Management Information Systems and digital learning assessments, education data ecosystems across the region remain profoundly fragmented. Critical data on early-grade learning outcomes, classroom pedagogy, and teacher performance is routinely collected in institutional silos and rarely shared across organisational boundaries. This fragmentation limits collaborative planning, generates redundant data collection efforts, and restricts the ability of education actors to target interventions toward the most vulnerable learner populations. A preliminary search of MEDLINE, the Cochrane Database of Systematic Reviews, and JBI Evidence Synthesis identified no current or underway systematic or scoping reviews specifically addressing data-sharing practices within foundational learning systems in SSA.

Objective

To systematically map and synthesise the extent, nature, and characteristics of evidence on data-sharing practices, including barriers, facilitators, interventions, and governance frameworks, among stakeholders within foundational learning systems across Sub-Saharan Africa.

Inclusion criteria

This review will include studies involving stakeholders embedded in foundational learning data systems (ministries of education, national statistical offices, schools, non-governmental organisations, development partners, education technology actors, and researchers) in SSA. Eligible sources must address barriers, facilitators, interventions, or governance frameworks relating to education data sharing in formal and non-formal foundational learning settings (Early Childhood Care and Education through Grade 5). All quantitative, qualitative, and mixed-methods study designs will be considered, alongside grey literature including policy documents and programme evaluations. Studies published in English or French from 2010 onwards will be included.

Methods

This review follows the Joanna Briggs Institute (JBI) methodology for scoping reviews and will be reported using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR). A three-step search strategy will be applied across seven electronic databases and grey literature sources. Two reviewers will independently screen all records at title and abstract stage and full-text stage, following a calibration pilot on 25 records requiring 75% inter-rater agreement before proceeding. Data will be extracted using a standardised matrix developed for this review. Analysis will combine descriptive numerical summary, thematic synthesis, and visual mapping. No formal critical appraisal will be undertaken, consistent with JBI guidance for scoping reviews. This protocol is registered on the Open Science Framework.

Introduction

Foundational literacy and numeracy represent the key bases of lifelong learning, individual economic opportunity, and sustainable socio-economic development (UNESCO, 2022; World Bank, 2023). In Sub-Saharan Africa (SSA), achieving universal foundational learning remains one of the most pressing educational challenges. The adoption of United Nations Sustainable Development Goal 4 (SDG 4) in 2015 marked a decisive global paradigm shift, moving the primary focus of educational systems from measuring school access and enrolment metrics to evaluating actual learning outcomes (United Nations, 2015; UNESCO, 2015). Realising these target outcomes requires evidence-informed decision-making driven by accurate, timely, and accessible education data (UNESCO, 2022).

Despite growing investments in Education Management Information Systems (EMIS) and digital learning assessments across SSA, education data ecosystems remain profoundly fragmented (Pambe et al., 2025b). Critical data points regarding early-grade learning outcomes, classroom pedagogy, teacher performance, and school infrastructure are frequently collected in isolated silos. Ministries of Education (MoEs), National Statistical Offices (NSOs), non-governmental organisations (NGOs), multilateral development partners, Educational Technology (EdTech) vendors, and academic researchers often operate their own distinct data collection pipelines (Pambe et al., 2025b). Consequently, valuable learning data is rarely shared, integrated, or cross-analysed across institutional boundaries. This fragmentation hinders comprehensive policy planning, leads to redundant data collection efforts, and restricts the ability of local educators and regional administrators to target interventions toward underrepresented or vulnerable learner populations, such as those in rural, pastoralist, or refugee-affected contexts (Pambe et al., 2025a).

To address these systemic inefficiencies, establishing safe, trustworthy, and interoperable data-sharing practices is essential. In the context of this review, foundational learning systems are defined as formal and non-formal educational structures encompassing Early Childhood Care and Education (ECCE) through upper primary schooling (Grades 1 to 5), where learners acquire core literacy, numeracy, and socio-emotional competencies. Data sharing practices refer to the institutional, technical, legal, and interpersonal processes through which educational data, including student assessment results, institutional metrics, and teacher performance indicators, is exchanged, integrated, and utilised across organisational boundaries.

The dynamics of data sharing within SSA education systems are shaped by a complex interplay of systemic factors. Barriers encompass structural and socio-technical friction that restricts data flow, such as inadequate digital infrastructure, erratic power supplies, legacy software architectures, restrictive legal mandates, zero-trust organisational cultures, and fears of institutional performance scrutiny or political misalignment (Pambe et al., 2025b). Facilitators represent elements that enable effective exchange, such as standardised data protocols, direct feedback loops, user-friendly analytical dashboards, clear data transfer agreements, and aligned donor-government financial incentives (UNESCO Institute for Statistics, 2023). Interventions and mechanisms include localised technological tools, offline digital platforms, automated data anonymisation engines, and structural capacity-building initiatives designed to operationalise secure data pipelines (EdTech Hub, 2023). Governance and policy frameworks comprise the formal legal instruments, regulations, ethical guidelines, and regional agreements, such as national privacy laws or alignment with international FAIR data principles (Wilkinson et al., 2016), that define data ownership, security, and usage boundaries.

While there is a growing body of literature examining educational administration, digital transformation, and EdTech innovations in low-resource settings (EdTech Hub, 2023; UNESCO, 2022), evidence specifically

addressing data sharing mechanisms in SSA's foundational learning sector remains widely dispersed across peer-reviewed journals, multilateral donor evaluations, grey literature reports, and governmental policy documents (Pambe et al., 2025a). A preliminary search of MEDLINE, the Cochrane Database of Systematic Reviews, and JBI Evidence Synthesis was conducted, and no current or underway systematic or scoping reviews on this specific topic were identified.

A scoping review is the most appropriate methodological approach for this study. Unlike a systematic review that evaluates the effectiveness of a single standardised intervention, a scoping review is designed to map the extent, range, and nature of evidence across an emerging, heterogeneous, and interdisciplinary field (Arksey and O'Malley, 2005; Peters et al., 2020). Given that literature on educational data governance in SSA spans diverse quantitative, qualitative, policy, and grey literature sources, a scoping review allows researchers to synthesise disparate operational definitions, map multi-stakeholder dynamics, categorise implemented governance mechanisms, and pinpoint critical empirical gaps (Munn et al., 2018; Tricco et al., 2018). The overarching objective is to systematically map and synthesise the extent, nature, and characteristics of evidence regarding data-sharing practices among stakeholders within foundational learning systems across Sub-Saharan Africa.

Review Questions

Overarching question: What is known about the barriers and facilitators to data sharing among stakeholders in foundational education systems in Sub-Saharan Africa, and what approaches have been proposed and/or implemented to promote trustworthy and safe data sharing practices?

Sub-questions:

1. What barriers prevent data sharing among stakeholders in foundational learning systems in SSA?
2. What facilitators influence and encourage data sharing among stakeholders in foundational learning systems in SSA?
3. What evidence-based interventions or technical mechanisms have been implemented or proposed to promote trustworthy, timely, and safe data sharing in foundational learning contexts?
4. What governance and policy frameworks exist to guide data sharing among stakeholders in foundational learning systems in SSA?
5. What gaps exist in the empirical and policy evidence on data sharing in foundational learning systems, particularly across different SSA administrative and geographical sub-regions?

Inclusion Criteria

Eligibility is defined using the Participants, Concept, Context (PCC) framework, consistent with JBI methodology for scoping reviews.

Participants

This review will consider studies and reports involving all key stakeholders embedded within foundational learning data systems. Eligible participants include national ministries of education (MoEs) and sub-

national education boards; national statistical offices (NSOs); primary school administrators, head teachers, and classroom teachers; non-governmental organisations, civil society organisations, and international development partners; EdTech providers, software developers, and private sector actors delivering learning solutions; and educational researchers and academic institutions analysing early-grade learning metrics.

Studies exclusively involving stakeholders in secondary education, higher or tertiary education, vocational training, or non-educational sectors without direct integration into foundational learning data systems will be excluded.

Concept

The core concept under investigation is data sharing practices. Eligible sources must address one or more of the following elements regarding early-grade literacy and numeracy data: structural, technical, legal, financial, or socio-cultural barriers to data exchange; mechanisms, incentives, or relational factors functioning as facilitators of data exchange; specific interventions, tools, or architectures (such as integrated EMIS platforms, offline data syncing mechanisms, open data portals, or automated privacy engines) aimed at securing or scaling data sharing; and formal governance structures, policies, regulations, and legal agreements governing data ownership, ethical handling, interoperability, and system trust.

Studies focusing exclusively on primary data collection methods or assessment tool validation without addressing data sharing, transmission, governance, or multi-stakeholder integration will be excluded.

Context

The review will focus on foundational learning systems (Early Childhood Care and Education through Upper Primary, Grades 1 to 5) situated within countries in Sub-Saharan Africa (SSA), as defined by the World Bank regional classification. This includes formal public and private schooling systems, community-based education programmes, government monitoring initiatives, regional learning assessment projects, and digital and EdTech deployment environments across urban, peri-urban, rural, pastoralist, and conflict-affected or emergency settings.

Studies conducted exclusively outside Sub-Saharan Africa will be excluded. Studies where the SSA setting is incidental rather than central to the research focus will also be excluded.

Types of Sources

This review will consider a broad range of study designs and literature types to capture the multifaceted landscape of educational data governance. Eligible sources include quantitative study designs (randomised and non-randomised controlled trials, interrupted time series, cohort studies, and descriptive cross-sectional surveys), qualitative designs (phenomenology, grounded theory, ethnography, action research, and case studies), mixed-methods designs, and existing systematic or scoping reviews (used only for reference harvesting, not as primary data sources). Grey literature including institutional programme evaluations, technical white papers, national policy documents, legal framework decrees, conference proceedings, dissertations, and official donor reports will also be included.

Fact sheets, newspaper articles, opinion blogs, commercial product advertisements, and editorial opinion pieces without empirical or analytical policy substance will be excluded.

Language of Publication

Literature published in English and French will be included to ensure balanced coverage of both Anglophone and Francophone administrative education systems across Sub-Saharan Africa. Studies published in other languages without an available English or French translation will be excluded. This restriction is acknowledged as a potential source of language bias and will be noted as a limitation in the final review.

Year of Publication

Studies published from 2010 onwards will be included. This lower boundary captures data-sharing practices in the years immediately preceding the 2015 ratification of United Nations Sustainable Development Goal 4 (SDG 4), which marked a systemic global transition toward monitoring foundational learning outcomes and catalysed modern EMIS development and digital data integration strategies across SSA, providing an empirical baseline against which post-2015 developments in data-sharing practice can be assessed. Studies published prior to 2010 are more likely to reflect substantially different technological and governance realities and are less likely to represent data-sharing practices relevant to current EMIS and digital data-sharing contexts.

Geographical Scope

Studies will be included if conducted within Sub-Saharan African settings encompassing both formal and non-formal foundational learning environments, with particular attention to contexts where data infrastructure is nascent, fragmented, or heavily donor-dependent, including rural, peri-urban, pastoralist, and refugee-affected areas. Studies will be excluded if the SSA setting is incidental rather than significant to the research focus, or if the context relates solely to higher education or health systems without direct linkage to learning outcomes.

Methods

This scoping review will be conducted in accordance with the Joanna Briggs Institute (JBI) methodology for scoping reviews (Peters et al., 2020). Findings will be reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) (Tricco et al., 2018). This protocol is registered on the Open Science Framework prior to commencing data collection.

Search Strategy

The search strategy aims to locate both published and unpublished evidence. A three-step approach will be employed.

In the first step, a preliminary limited search will be conducted in ERIC (searched via the ProQuest interface where institutional access is available, or otherwise via the free eric.ed.gov / api.ies.ed.gov public interface, which indexes the same underlying database) and Education Source (EBSCO) to identify relevant articles on the topic. Text words in titles and abstracts and index terms assigned by these databases will be analysed to develop a comprehensive keyword matrix and Boolean search string.

In the second step, the identified keywords and Boolean strings will be adapted and applied across seven electronic databases: ERIC (via the ProQuest interface where institutional access is available, or the free eric.ed.gov / api.ies.ed.gov interface where it is not), Education Source (EBSCO), the British Education

Index, Scopus, Web of Science Core Collection, Google Scholar (first 200 relevant results, or a defensible relevance-ranked slice where the on-topic hit rate is unusually high), and OpenAlex (a free, open, API-driven scholarly index used to extend citation coverage where institutional access to Scopus or Web of Science is unavailable). Education Source, the British Education Index, Scopus, and Web of Science Core Collection require institutional subscription access; where this is not available to the review team, the gap will be documented transparently in the review's search log, and OpenAlex and Google Scholar will serve as the review's open-access substitutes for citation-index coverage. These databases were selected to provide broad coverage of education, social sciences, policy, and African-produced research.

In the third step, the reference lists of all included sources and relevant reviews discovered during screening will be manually screened to identify additional eligible items. Key grey literature repositories will also be hand-searched, including the UNESCO International Institute for Educational Planning (IIEP) open archive, the World Bank Open Knowledge Repository, the Association for the Development of Education in Africa (ADEA), the Unlocking Data Initiative's own Evidence Library (unlockingdata.africa), and the African Journals Online (AJOL) network. AJOL's own search interface disallows automated, scripted querying under its robots.txt, but is directly reachable through a standard browser session; where reachable this way it will be queried directly, with the top relevance-ranked page(s) fully screened given its very large, effectively unbounded result counts for broad multi-term queries. Where only scripted access is available to support the hand search, a site-restricted web search (via DuckDuckGo, since Google returns a CAPTCHA challenge on repeated site-restricted queries and Bing silently drops both site: filters and quoted phrase terms) will substitute for AJOL's own search tool.

The search will be restricted to English and French language publications from 2010 to the present. All retrieved records will be imported into Zotero for centralised reference management and deduplication. The full search strategy for each database, including all terms, Boolean operators, and applied limits, will be reported in Appendix I and in the final review.

Sample search string

The search strategy is structured around the three PCC elements, translated into three Boolean blocks combined using AND operators. The sample search string below is adapted for ERIC (via ProQuest or the free eric.ed.gov / api.ies.ed.gov interface, see above). Strings for all other databases will be adapted to their indexing structures and documented in full in Appendix I. Reviewers should note that ERIC's own search interface, whether accessed via ProQuest or its free public API, performs loose, word-level matching within indexed description fields rather than strict phrase-level Boolean logic; a single literal Boolean string combining several multi-word phrase blocks may therefore return results that share little topical relevance with the query as written, and the search should be supplemented with a series of narrower, single-concept-pair queries rather than relying on the yield of the full literal string alone.

("data sharing" OR "data exchange" OR "data governance" OR "information sharing" OR "data interoperability" OR "open data" OR "data integration" OR "education data" OR "educational data" OR "school data" OR "learning data" OR "data ecosystem" OR "data access" OR "EMIS" OR "education

management information system" OR "education management information systems") AND (Africa OR African OR "Sub-Saharan" OR Cameroon OR Kenya OR Nigeria OR Ghana OR Tanzania OR Uganda OR Senegal OR Ethiopia OR Malawi OR Zambia OR Zimbabwe OR Rwanda OR Mozambique OR "developing countries" OR "low-income countries") AND (education OR school OR learning OR literacy OR numeracy OR "foundational learning" OR "primary education" OR "early grade" OR "basic education")

Limits applied: Language: English, French. Date: 2010 to present.

Study Selection

Following export of all search results and automated deduplication in Zotero, supplemented by manual line-by-line review to remove residual duplicates, records will proceed through a two-stage independent screening process.

Prior to formal screening, a calibration pilot will be completed. A random sample of 25 titles and abstracts will be screened independently by all reviewers against the predefined PCC inclusion criteria. An inter-rater agreement threshold of 75% must be achieved before proceeding to full screening. If this threshold is not met, the eligibility definitions will be refined and a second pilot conducted on a separate sample.

At the first stage, two reviewers will independently screen all deduplicated records at title and abstract level against the PCC eligibility criteria. Records that clearly do not meet criteria will be excluded. Records that appear relevant or where eligibility is uncertain will be retained for full-text review.

At the second stage, full texts of all retained records will be retrieved and assessed independently by two reviewers in JBI SUMARI. Each record will be assessed against the full set of PCC inclusion criteria. The explicit reason for exclusion will be recorded for each excluded full-text document.

Any disagreements at either stage will be resolved through discussion and consensus. If consensus cannot be reached, a fourth, senior reviewer will make the final determination. The complete study flow will be presented using a PRISMA-ScR flow diagram in the final review.

Data Extraction

Data extraction will be independently performed by two reviewers using the standardised extraction matrix detailed in Appendix II. Prior to full extraction, the tool will be pre-tested on three heterogeneous included sources: one peer-reviewed journal article, one donor-funded programme evaluation, and one national policy document. Modifications arising from pre-testing will be documented and agreed by the full team.

Any discrepancies in extraction between reviewers will be resolved through discussion and cross-checking against the primary source. Where necessary, authors of primary papers or institutional reports will be contacted by email up to two times over a 14-day window to request missing data, clarify unclear figures, or confirm specific configurations.

In accordance with JBI scoping review guidelines, formal critical appraisal of individual sources will not be undertaken. The primary objective is to map the volume, nature, and characteristics of the existing literature rather than evaluate methodological quality.

Data Analysis and Presentation

The gathered data will undergo descriptive, thematic, and visual synthesis to provide a structured map of the educational data-sharing ecosystem in SSA. The analysis will directly address the overarching review objective and each of the five sub-questions.

Descriptive numerical summary

Quantitative data will be summarised using frequencies and counts to describe the distribution of included sources by year of publication, country or sub-region, type of publication, study design, stakeholder group, and sub-question focus. This summary will identify patterns in the existing literature and highlight where evidence is concentrated or absent.

Thematic synthesis

Qualitative and mixed-methods data will be coded inductively and deductively using themes aligned to the five sub-questions: barriers, facilitators, interventions and mechanisms, governance and policy frameworks, and evidence gaps. Coding will be conducted in MAXQDA. Thematic categories will be developed iteratively as coding progresses, with findings reviewed collectively by the team at key analytical milestones.

Visual presentation

Three visual outputs will be produced to support interpretation. First, a geospatial map or regional matrix will plot the frequency of data-sharing literature across SSA countries, explicitly identifying where evidence is concentrated and where empirical voids exist. Second, a stakeholder thematic matrix will cross-tabulate identified barriers and facilitators against specific stakeholder groups, highlighting points of tension between macro-level ministry data goals and school-level implementation realities. Third, an intervention typology matrix will group implemented or proposed interventions by category (technical, legal, or capacity-building) and by operational status (proposed versus formally implemented).

A descriptive narrative summary will accompany all visual outputs, organising findings sequentially by sub-question and contrasting data-sharing dynamics across SSA sub-regions and governance contexts. The synthesis will highlight how institutional trust operates as a critical variable in data-sharing behaviour and identify the structural, political, and technical gaps in the current evidence base.

Projected Timeline

- May to June 2026: Protocol finalisation and OSF registration prior to commencing searches.
- June to July 2026: Database searches across all seven academic databases and grey literature sources.
- July 2026: Calibration pilot and title and abstract screening.
- August 2026: Full-text screening with reasons for exclusion recorded.
- August to September 2026: Data extraction with pre-testing on three sources.
- September to October 2026: Thematic and descriptive synthesis in MAXQDA.
- October 2026: Draft report and Workshop 3 peer feedback.
- October to November 2026: Final manuscript revision, submission, and CoP plenary presentation.

Deviations from Protocol

This protocol sets out the planned methodology to ensure transparency, rigour, and reproducibility. All methods will be adhered to in line with JBI guidance. Where adjustments become necessary in response to the nature or volume of retrieved evidence, these will be carefully documented, clearly justified, and reported transparently in the final review under a dedicated Deviations from Protocol subsection.

Potential deviations may include the inclusion of additional grey literature sources identified as highly relevant during the review, refinement of the thematic coding framework in response to unexpected patterns in the evidence, or minor adjustments to eligibility criteria to improve clarity. No deviations are anticipated at the protocol stage. Any decision to impose a date restriction narrower than 2010 to present, or to modify the language restrictions, would require explicit justification and full documentation before implementation.

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Appendix I: Full Search Strategy

The full search strategy actually run for each database is presented below, including all Boolean operators, quoted phrases, and applied limits, starting with ERIC (via the ProQuest interface where institutional access is available, or the free eric.ed.gov / api.ies.ed.gov interface where it is not). Where a database's own search does not reliably honour phrase or Boolean logic, this is noted alongside its string rather than left implicit.

ERIC (ProQuest, or the free eric.ed.gov / api.ies.ed.gov interface) - Full Search String

("data sharing" OR "data exchange" OR "data governance" OR "information sharing" OR "data interoperability" OR "open data" OR "data integration" OR "education data" OR "educational data" OR "school data" OR "learning data" OR "data ecosystem" OR "data access" OR "EMIS" OR "education management information system" OR "education management information systems") AND (Africa OR African OR "Sub-Saharan" OR Cameroon OR Kenya OR Nigeria OR Ghana OR Tanzania OR Uganda OR Senegal OR Ethiopia OR Malawi OR Zambia OR Zimbabwe OR Rwanda OR Mozambique OR "developing countries" OR "low-income countries") AND (education OR school OR learning OR literacy OR numeracy OR "foundational learning" OR "primary education" OR "early grade" OR "basic education")

OpenAlex (api.openalex.org) - Full Search String

filter=title_and_abstract.search:("data sharing" OR "data exchange" OR "data governance" OR "information sharing" OR "data interoperability" OR "open data" OR "data integration" OR "education data" OR "educational data" OR "school data" OR "learning data" OR "data ecosystem" OR "data access" OR EMIS OR "education management information system") AND (Africa OR African OR "Sub-Saharan") AND (education OR school OR learning OR literacy OR numeracy OR "foundational learning" OR "primary education" OR "early grade" OR "basic education"), publication_year:2010-2026, sort=relevance_score:desc, per_page=200

Google Scholar (scholar.google.com) - Full Search Strings

English: "education management information system" Africa ("data sharing" OR "data governance") school

French (variant 1): "partage de donnees" education Afrique ecole primaire

French (variant 2): "systeme d'information de gestion de l'education" Afrique donnees

World Bank Open Knowledge Repository (openknowledge.worldbank.org) - Full Search Strings

"education management information system"

EMIS education data governance

"data governance" "foundational learning"

"data sharing" "primary education" Africa

"data interoperability" education

ADEA (adeanet.org) - Full Search Strings

*“education management information system”
data sharing education governance*

AJOL (ajol.info) - Full Search Strings

*“education management information system” “data sharing”
“education management information system”*

UNESCO IIEP open archive (iiep.unesco.org) - Search Attempted, Native Tooling Unreliable

“education management information system” and related EMIS/data governance terms, plus the “Data and information systems” topic facet on the Publications catalogue

Unlocking Data Initiative Evidence Library (docs.unlockingdata.africa) - Full Search String

“data sharing governance”

Appendix II: Data Extraction Form

The following form will be transferred into an electronic spreadsheet for use during extraction. It will be pre-tested on three heterogeneous sources before full extraction begins. One completed form per included source. Modifications agreed during pre-testing will be documented.

Extraction category	Field label	Guidance notes and target variables
Administrative	Source metadata	Record author(s), exact year, source type (peer-reviewed article, white paper, programme evaluation, policy document, dissertation), and funding source.
Structural	Setting and scale	Country, region, or sub-national locality; urban, rural, peri-urban, or pastoralist; scale of the data system (national, regional, district, or institutional pilot).
System typology	Educational context	Level of foundational learning targeted (ECCE, Lower Primary Grades 1 to 3, Upper Primary Grades 4 to 5); type of institutions (public, private, community-based).
Human infrastructure	Stakeholders involved	List all actors participating in the data exchange network described (for example: NSOs, MoE officials, teachers, external donors, EdTech developers, NGOs, researchers).
GESI	Gender, equity, and social inclusion dimensions	Does the source disaggregate data by gender? Yes / No / Partial. Does it address disability or marginalised populations? Yes / No. Are intersecting vulnerabilities examined (for example, girls in rural or conflict-affected areas)? Yes / No / Describe.
Sub-question 1	Barriers to sharing	Extract explicit text regarding technical friction (lack of electricity, legacy databases), legal limits (absent privacy laws), cultural blocks (zero-trust environments), or resource deficits.

Sub-question 2	Facilitators of sharing	Document factors that improved data workflows: direct feedback loops, user-friendly dashboards, clear legal safety nets, performance-linked financial incentives, inter-agency trust mechanisms.
Sub-question 3	Interventions and mechanisms	Identify specific technical tools or processes designed to facilitate trustworthy exchange: localised dashboards, offline data collection tools, automated anonymisation software, capacity-building initiatives.
Sub-question 4	Governance and policies	Track formal frameworks: alignment with international FAIR data principles, compliance with national privacy laws, institutional data-sharing MoUs, data transfer agreements, regional regulatory frameworks.
Sub-question 5	Gaps and limitations	Transcribe explicit statements from source authors regarding research limitations, operational gaps, or structural constraints specific to the local context.
Synthesis	Key findings and recommendations	Principal conclusions and implications for policy, practice, and future research as stated by the authors.